WORKED SOLUTION

STEP I 1991, Question 2

Pure • Rates of Change

STEP I 1991, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s1-q2

The key idea

Write the melting law in terms of the radius. For a sphere, $\frac{\mathrm{d}V}{\mathrm{d}t} = 4\pi r^2 \frac{\mathrm{d}r}{\mathrm{d}t}$, and the surface area is $4\pi r^2$ too, so "volume lost at a rate proportional to surface area" says simply that the radius shrinks at a constant rate. Both snowballs share the constant, so both radii fall by the same amount in any time. Everything then becomes a function of one variable, the amount x by which each radius has shrunk.

Solution

The radii shrink at the same constant rate

Let a snowball have radius r, volume $V = \frac{4}{3}\pi r^3$ and surface area $S = 4\pi r^2$. The melting law is $\frac{\mathrm{d}V}{\mathrm{d}t} = -kS$ for a constant $k > 0$, the same for both snowballs. By the chain rule,

$$4\pi r^2 \frac{\mathrm{d}r}{\mathrm{d}t} = -k \cdot 4\pi r^2 \quad \Longrightarrow \quad \frac{\mathrm{d}r}{\mathrm{d}t} = -k,$$

dividing by $4\pi r^2$, which is legitimate while the snowball still exists ($r > 0$). So after time t each radius has decreased by the same amount $x = kt$: the head has radius $2R - x$ and the body has radius $3R - x$, valid for $0 \le x \le 2R$ (the head disappears at $x = 2R$).

Volume at half height

Frosty's height is the sum of the two diameters, since the head rests on top of the body:

$$h = 2(2R - x) + 2(3R - x) = 10R - 4x.$$

Initially $h = 10R$, so half height means $10R - 4x = 5R$, that is $x = \frac{5}{4}R$. This is less than $2R$, so the head is still there, and the radii are then $\frac{3}{4}R$ and $\frac{7}{4}R$. The factor $\frac{4}{3}\pi$ cancels

in the ratio, so

$$\frac{V}{V_0} = \frac{\left(\frac{3}{4}\right)^3 + \left(\frac{7}{4}\right)^3}{2^3 + 3^3} = \frac{(27+343)/64}{35} = \frac{370}{2240} = \frac{37}{224}.$$

So at half his height Frosty has only $\frac{37}{224} \approx 0.165$ of his original volume: volume scales like length cubed.

The maximum of $\mathrm{d}V/\mathrm{d}S$

Write $a = 2R - x$ and $b = 3R - x$. For $0 \le x < 2R$,

$$V = \tfrac{4}{3}\pi(a^3 + b^3), \qquad S = 4\pi(a^2 + b^2),$$

and since $\frac{\mathrm{d}a}{\mathrm{d}x} = \frac{\mathrm{d}b}{\mathrm{d}x} = -1$,

$$\frac{\mathrm{d}V}{\mathrm{d}x} = -4\pi(a^2 + b^2), \qquad \frac{\mathrm{d}S}{\mathrm{d}x} = -8\pi(a + b).$$

Ask yourself V and S are both functions of time (or of x). What does $\mathrm{d}V/\mathrm{d}S$ mean?

As x increases, S strictly decreases (because $\frac{\mathrm{d}S}{\mathrm{d}x} < 0$), so V can be regarded as a function of S. By the chain rule $\frac{\mathrm{d}V}{\mathrm{d}S} = \frac{\mathrm{d}V}{\mathrm{d}x} \Big/ \frac{\mathrm{d}S}{\mathrm{d}x}$:

$$\frac{\mathrm{d}V}{\mathrm{d}S} = \frac{a^2 + b^2}{2(a+b)}.$$

Now use $b - a = R$, which never changes. Put $s = a + b = 5R - 2x$. Then $a^2 + b^2 = \frac{1}{2}((a+b)^2 + (b-a)^2) = \frac{1}{2}(s^2 + R^2)$, so

$$\frac{\mathrm{d}V}{\mathrm{d}S} = \frac{s^2 + R^2}{4s} = \frac{s}{4} + \frac{R^2}{4s} = f(s), \text{ say.}$$

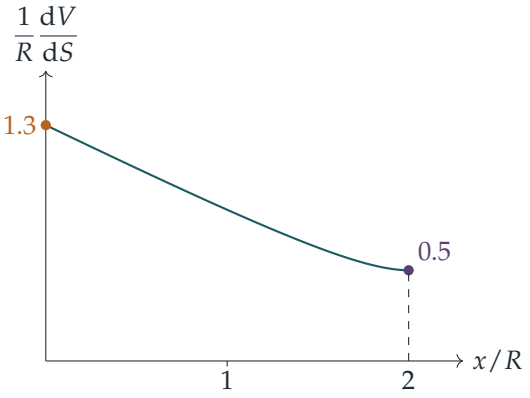
As x runs from 0 to $2R$, s runs from $5R$ down to R. Now

$$f'(s) = \frac{1}{4} - \frac{R^2}{4s^2} = \frac{s^2 - R^2}{4s^2} > 0 \quad \text{for } s > R,$$

so f is increasing on $[R, 5R]$, and its largest value is at $s = 5R$, that is, at the very start ($x = 0$):

$$\left(\frac{\mathrm{d}V}{\mathrm{d}S}\right)_{\max} = \frac{5R}{4} + \frac{R}{20} = \frac{13R}{10}.$$

(Directly: $\frac{4R^2 + 9R^2}{2 \cdot 5R} = \frac{13}{10}R$.) The value then falls steadily to $f(R) = \frac{1}{2}R$ just as the head vanishes.



Checks

- A step-by-step simulation of $\frac{dV}{dt} = -kS$ for each ball (with $R = k = 1$) reaches half height at $t \approx 1.25$, with volume ratio 0.16518, matching $\frac{37}{224} = 0.165178\ldots$.
- Finite differences of the simulated V and S agree with $\frac{a^2+b^2}{2(a+b)}$ throughout and never exceed $1.3R$.
- Units: dV/dS is a volume over an area, so a length, as $\frac{13}{10}R$ is.

Common slips

- Taking the height as the sum of the radii, not of the diameters. (Here it happens not to change x, but it is wrong.)
- Computing V/S instead of dV/dS; for a single sphere $V/S = r/3$ but $dV/dS = r/2$.
- Hunting for a stationary point: the turning point of f is at $s = R$, the endpoint, so the maximum is at the other end of the interval.

Beyond STEP

The heart of the question is that a "rate proportional to surface area" law turns into a constant rate for the radius, a first example of how geometry can make a differential equation trivial. The second half is a parametric derivative, dV/dS along a curve $(S(t), V(t))$, and the care needed to find a maximum on a closed interval, where it may sit at an endpoint rather than at a stationary point; Bartle and Sherbert's *Introduction to Real Analysis* explains why such maxima must exist. For modelling with differential equations, try Boyce and DiPrima, *Elementary Differential Equations*.

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